Class12

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```
url <- "https://bioboot.github.io/bggn213_W19/class-material/rs8067378_ENSG00000172057.6.txt"
dat <- read.table(url, header = TRUE, stringsAsFactors = TRUE)
```

```
head(dat)
```

```
   sample geno      exp
1 HG00367  A/G 28.96038
2 NA20768  A/G 20.24449
3 HG00361  A/A 31.32628
4 HG00135  A/A 34.11169
5 NA18870  G/G 18.25141
6 NA11993  A/A 32.89721
```

```
str(dat)
```

```
'data.frame':   462 obs. of  3 variables:
 $ sample: Factor w/ 462 levels "HG00096","HG00097",..: 170 424 165 37 299 219 96 285 138 17
 $ geno  : Factor w/ 3 levels "A/A","A/G","G/G": 2 2 1 1 3 1 2 1 3 2 ...
 $ exp   : num  29 20.2 31.3 34.1 18.3 ...
```

```
table(dat$geno)
```

```
A/A A/G G/G
108 233 121
```

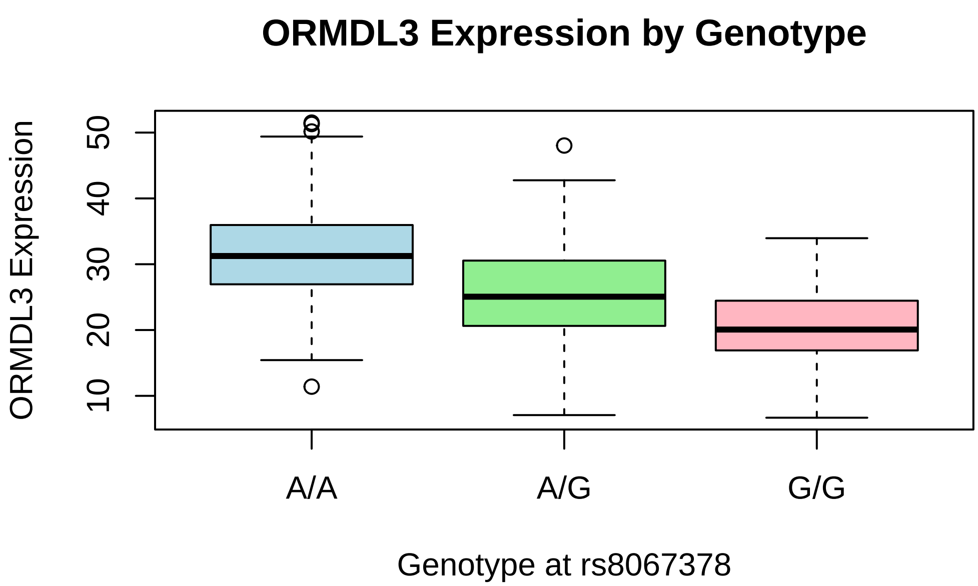
Q13.

The three genotypes show clear differences in ORMDL3 expression. The median expression for individuals with the A/A genotype is 31.24847, which is the highest among the groups. The A/G heterozygotes show an intermediate median expression of 25.06486, while the G/G genotype has the lowest median expression at 20.07363. This pattern suggests a stepwise decrease in ORMDL3 expression as the number of G alleles increases.

```r
tapply(dat$exp, dat$geno, median)
```

```
     A/A      A/G      G/G 
31.24847 25.06486 20.07363 
```

```r
boxplot(exp ~ geno, data = dat,
        xlab = "Genotype at rs8067378",
        ylab = "ORMDL3 Expression",
        main = "ORMDL3 Expression by Genotype",
        col = c("lightblue", "lightgreen", "lightpink"))
```



Q14.

The boxplot comparing A/A, A/G, and G/G genotypes shows a clear downward trend in ORMDL3 expression from A/A to G/G. Because individuals with the G/G genotype express ORMDL3 at substantially lower levels than those with A/A, the data indicate that rs8067378 does influence ORMDL3 expression. In other words, this SNP acts as an eQTL, with the G allele associated with reduced expression of the gene.